

Programming Project Report

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Problem Statement:

The goal of the assignment is to give students experience with user interaction and OpenGL callback functions. The task for the students was to create and design a custom Pacman game.

Design:

So, I decided to start the project by with my project 1 code. However, before I started, I basically removed everything besides the underlying functions I'll need like main, display, init, etc. But most of the code inside was deleted and re-done as required. I decided that Pacman, Food and Ghosts will all have their own classes. This helped to separate everything and sense all three of these classes had to be a little bit different from each other. I had a recommendation from AI to use the timer function, which is apparently a little better than throwing everything in display, so most of my work (besides drawing) is inside that timer function. The pro of this design was definitely a nice workflow, and it was easier to read the proj2.cpp. The con of this was probably slowing down to making sure classes were correctly made and it was a bit tedious.

Implementation:

I worked my way down the assignment page. Completing one task at a time. So initially I spent my time creating classes for each of the 3 main objects (Pacman, Food, and Ghosts), and then I gave each of them a draw function to draw each of them uniquely. I then started working on having the food and ghosts moved around the screen in random directions. I had some fun and added a random function where the ghosts will change directions randomly (1/1000 chance, but it's in the display function so it happens somewhat often). I have the food be the slowest object, with Pacman being slightly faster and ghosts being faster than both. I felt like this was a good balance. With the food and ghosts done, I added keyboard controls for Pacman. Users can use the arrow keys to move Pacman around in any direction. Lastly, I did some heavy edits to the code to make the game actually function like a game. Adding logic when Pacman 'touches' food or ghosts. In my game, Pacman only has one life, so if you get touched by a ghost once, it ends the program. The user can collect points by eating food, 1 point for every food eaten. Once the user gets to 100 points, the user will see a special victory screen and all ghosts are removed so the user can eat all the food they want in peace. I started working on this code Friday morning. Yeah, I know I shouldn't do that, but I got behind with 3 other assignments all due Thursday. So, I worked on the project all Friday, throughout classes and as I am writing this at 9:40 pm.

AI Prompts:

1. How to make it where pacman can move diagonally? And have him rotate in the direction of travel...?
2. How to make it so diagonal isn't quicker than normal?
3. The rotation is not working still?
4. How to end the game safely?
5. Can we make it not seg fault to end?
6. How to make the background (black rn) go rainbow when points are ≥ 100
7. Can we put victory text on the screen?

Testing:

I tested the program by running the code several times, editing the win condition (I don't have time to get to 100 points to win) and looking out for bugs and inconsistencies. There really wasn't any inputs I used, just run the code like normal (run the MakeFile). I tested both endings, hitting a ghost and winning the game, both acted normally on my machine. Everything appeared to work as expected on my machine. The game closes when you touch a ghost, the background and text appear when you win the game. Food respawns (or really just moves to a random spot on the screen) when eaten so users can get to 100 points.

Figure 1: Start of Game



Figure 2: Moving Around 1

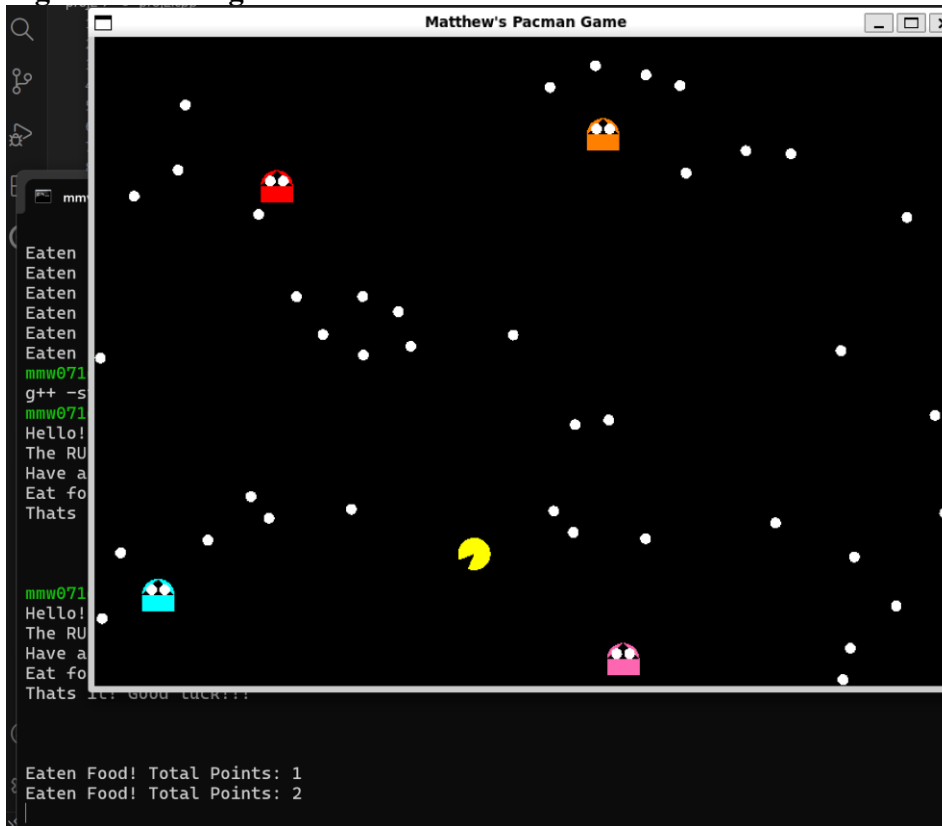


Figure 3: Moving Around 2

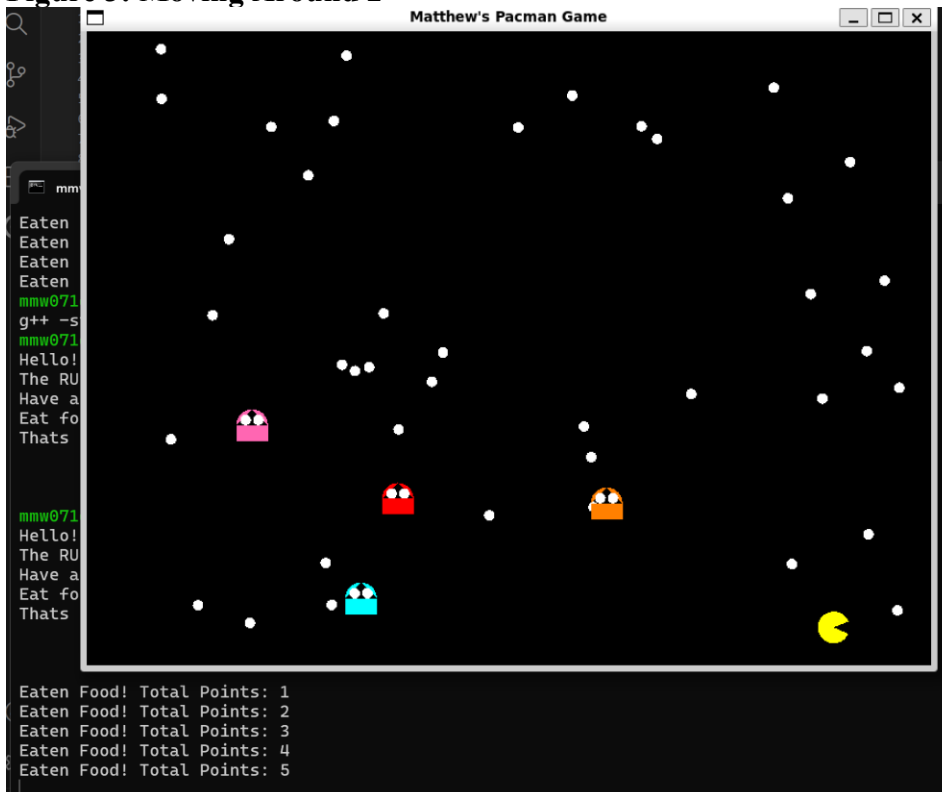


Figure 4: Moving Around 3

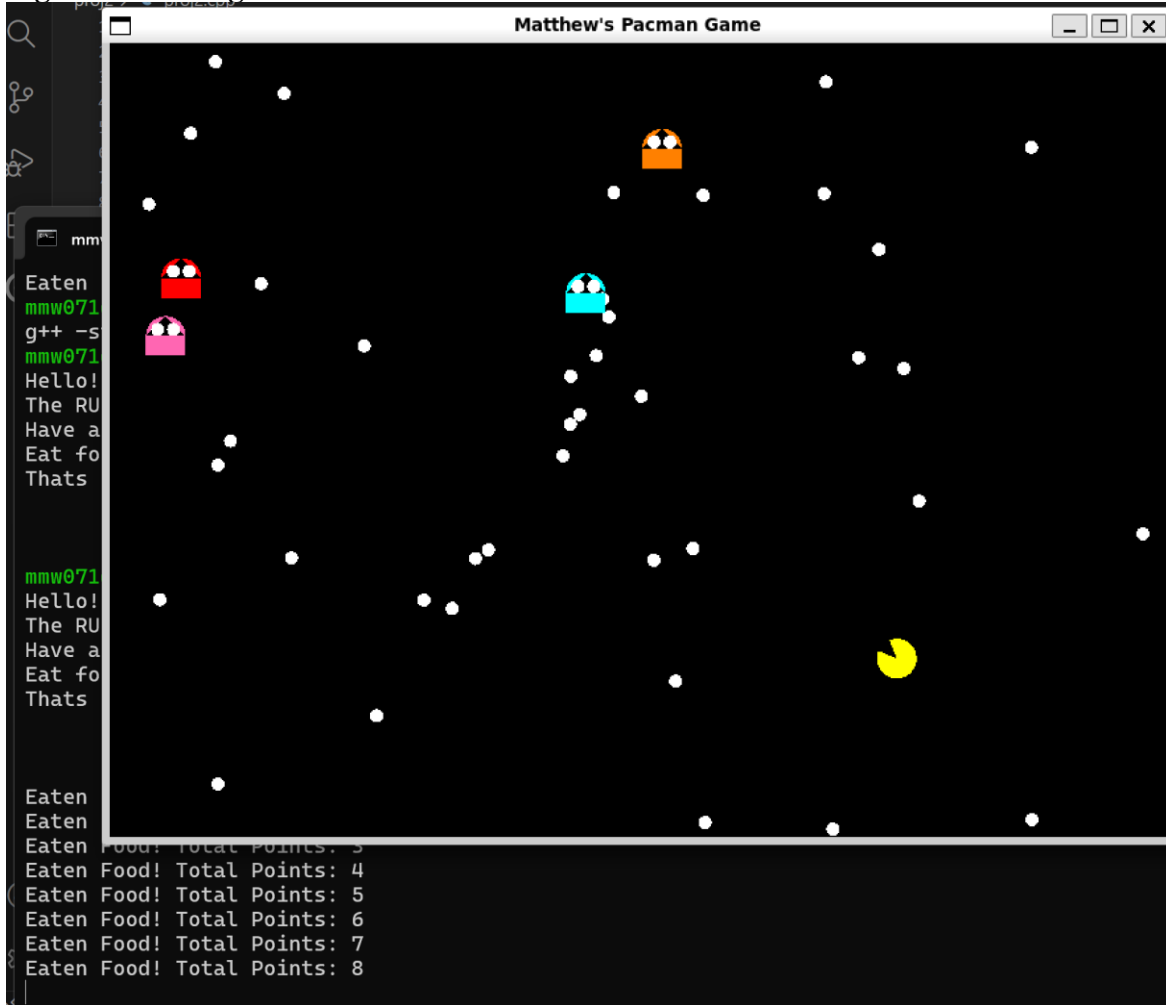


Figure 5: Pacman Hits Ghosts

```
mmw071@BOOK-13EP48IKF6:/mnt/c/Users/mww61/git/graphics/proj2$ ./proj2
Hello! Welcome to completely normal pacman.
The RULES are simple!
Have a ghost touch you, you die (pacman) and the program ends forever (not really).
Eat food to increase your score. Get to 100 points to get a special victory screen!
Thats it! Good luck!!!

Eaten Food! Total Points: 1
Eaten Food! Total Points: 2

OMG! A ghost got you! Pacman has died.
How could you have done this to him?
Game ENDING... :(
^[[D^[[Dmmw071@BOOK-13EP48IKF6:/mnt/c/Users/mww61/git/graphics/proj2$
```

Figure 6: Pacman Wins



Conclusions:

I believe the result of the assignment was successful, it produced a custom Pacman game with my own spin and fun. I think next time, I will try to start the project earlier, like when the project is released so I don't have to spend all day on Friday to finish the project on time. But, let's be honest, I'll probably do it again. As I stated above, it took me all day Friday, from morning to 9:56 pm, as I am writing this final sentence.